Global Fuel Prices & Taxation Dashboard: Project Report

Suramya Shrivastava

1. Introduction

1.1 Project Overview

This project is an interactive Tableau dashboard that studies global fuel prices and taxation from January 2020 to April 2026. The dataset holds 27,468 weekly records for 84 countries in 7 regions, with 10 fields: date, country, region, income level, subsidy level, petrol, diesel and LPG prices (USD per litre), Brent crude price and tax percentage.

The dashboard contains nine visualisations in three scenario sections. Each section answers the questions of one stakeholder: a policy consultant (Dr. Sarah Mitchell), a supply chain manager (Marcus Thompson) and an energy economics researcher (Prof. Elena Rodriguez). The work also includes a Tableau story, a project website and a GitHub repository.

1.2 Purpose

The purpose is to turn a large fuel dataset into clear, visual answers that support decisions. Policy makers can compare tax and subsidy approaches, logistics teams can plan fuel procurement and manage price risk, and researchers can study how income level relates to fuel tax, subsidies and prices.

2. Ideation Phase

2.1 Problem Statement

- **PS-1 (Dr. Sarah Mitchell, energy policy consultant):** I am trying to design fair and sustainable fuel tax and subsidy policies, but I cannot easily compare how countries and regions tax and subsidise fuel, because the data is spread across 84 countries, several fuel types and many years. This makes me feel uncertain and unable to back my recommendations with evidence.

- **PS-2 (Marcus Thompson, global supply chain manager):** I am trying to cut fuel costs and plan procurement across continents, but I cannot see which regions are cheapest or how volatile prices are, because the data is not organised by region, over time or with a forecast. This makes me anxious about sudden cost spikes.
- **PS-3 (Prof. Elena Rodriguez, energy economics researcher):** I am trying to find out how income level relates to fuel taxes, subsidies and prices, but I cannot quickly tell whether wealthier countries tax fuel differently, because patterns across income groups are hidden in thousands of records. This makes me curious but cautious.

2.2 Empathy Map Canvas

An empathy map was created for each persona.

- **Dr. Sarah Mitchell.** Says: which countries tax fuel the most, and do subsidies keep prices down. Thinks: tax rules differ so much that fair comparison is hard. Does: compares tax rates and subsidy levels and prepares recommendations. Feels: under pressure to justify every number.
- **Marcus Thompson.** Says: where diesel is cheapest and where prices are heading. Thinks: buying at the right time and place can cut costs. Does: compares regions, tracks trends and plans procurement. Feels: anxious about sudden price spikes.
- **Prof. Elena Rodriguez.** Says: do richer countries tax fuel more. Thinks: findings must be solid enough for peer review. Does: compares tax, subsidy and price by income group. Feels: curious but cautious about weak conclusions.

2.3 Brainstorming

Nine visualisation ideas were listed and grouped by scenario. Group 1 (policy): a tax map, a tax and subsidy comparison by region, and a subsidy versus price chart. Group 2 (logistics): a diesel trend with forecast, a top and bottom 10 market ranking, and a volatility heat map. Group 3 (research): tax by income level, subsidy mix by income level, and prices by fuel type and income. All three groups were rated high in importance and feasibility, and all nine ideas were selected.

3. Requirement Analysis

3.1 Customer Journey Map

The journey of a stakeholder using the dashboard has three stages.

- **Stage 1, Discover and Load Data:** the user opens the dashboard, reads the scenario and selects year and region filters. They feel curious but a little overwhelmed.
- **Stage 2, Explore and Visualise:** the user studies maps, trends and heat maps and clicks a region to filter other charts. They feel engaged as patterns appear.
- **Stage 3, Interpret and Decide:** the user notes patterns, shares findings and uses them for policy, procurement or research. They feel confident and ready to act.

3.2 Solution Requirements

Functional requirements

- Load the fuel prices CSV into Tableau.
- Check the date field and missing values, and create calculated fields (Subsidy Score, Avg Fuel Price, Diesel Volatility).
- Calculate average tax, average prices, subsidy-level comparisons, volatility and diesel forecasts by region, country, year and income level.
- Provide nine visualisations on one dashboard.
- Allow filtering by year and region, click-to-filter and tooltips.
- Allow updates by reloading a newer CSV, and share through Tableau Public, a website and GitHub.

Non-functional requirements: usability (clear titles, consistent colours), security (public, non-personal data only), reliability (consistent calculation rules), performance (27,468 rows filter smoothly), availability (online without installation) and scalability (larger CSVs with the same columns can be loaded).

3.3 Data Flow Diagram

The data flows through four processes. First, the data source (the CSV file) sends raw weekly data to process 1.0, Load and Validate Data. The validated records go to process 2.0, Clean Data and Create Calculated Fields, which reads and stores data in the Fuel Prices Dataset (D1). The prepared data goes to process 3.0, Build 9 Visualisations, which saves the Tableau workbook (D2). Finally, process 4.0, Assemble and Publish Dashboard, loads the workbook and sends the published dashboard to the stakeholders, who return filter and drill-down requests.

[Insert the DFD Level 0 diagram from the Data Flow Diagram template]

3.4 Technology Stack

- **Programming and calculations:** Tableau calculated fields (CASE, STDEV, AVG); Python (pandas) for checking and summarising data; HTML, CSS and JavaScript for the project website.
- **Database:** a single CSV file (global_fuel_prices_Dataset.csv).
- **Data ingestion:** Tableau data connection (Text file).
- **Transformation:** Tableau Desktop calculated fields, filters and sorting.
- **Visualisation:** Tableau Desktop and Tableau Public; Chart.js for the website.
- **Front end:** Tableau Public dashboard and the HTML website.

4. Project Design

4.1 Problem Solution Fit

The customers are policy consultants, logistics managers and researchers. Their jobs are to compare fuel taxes, check whether subsidies lower prices, track diesel trends and risk, and relate income level to tax and price. The root cause of the problem is that the data is large and varies by country, region, fuel type and time, so patterns stay hidden. Existing options are spreadsheets, static reports and price websites, which are hard to compare and not interactive. The solution is an interactive Tableau dashboard that moves users from feeling overwhelmed to feeling confident and in control.

[Insert the completed Problem Solution Fit canvas]

4.2 Proposed Solution

The approach is to load the CSV into Tableau, create three calculated fields, build nine worksheets grouped by scenario, combine them into one dashboard with filters, and publish it with a story, a website and a GitHub repository. Key features are the nine visualisations in three scenario sections, year and region filters with click-to-filter, a diesel forecast, top and bottom 10 markets, a volatility heat map, and income-level comparisons.

4.3 Solution Architecture

The architecture has seven components: the data source (CSV), the data connection (Tableau Desktop), data preparation (calculated fields), visual analytics (nine worksheets), the dashboard (three scenario sections with filters), publishing (Tableau Public, project website and GitHub) and the end users.

[Insert the solution architecture diagram]

5. Project Planning & Scheduling

5.1 Project Planning

The work was planned as 15 user stories in four sprints with story points assigned to each.

- **Sprint-1 (18 points):** load the dataset, tax map, tax and subsidy comparison by region, and subsidy versus price chart.
- **Sprint-2 (16 points):** diesel trend and forecast, top and bottom 10 markets, volatility heat map, and tax by income level.
- **Sprint-3 (14 points):** subsidy mix by income, prices by fuel type and income, dashboard filters and interaction, and publishing to Tableau Public.
- **Sprint-4 (13 points):** project templates and documentation, demo video, and the GitHub repository and website.

6. Functional and Performance Testing

6.1 Performance Testing

- **Data rendered:** 27,468 rows and 10 fields loaded from the CSV (84 countries, 7 regions, 6 January 2020 to 6 April 2026).
- **Data preprocessing:** the date field is set to Date type, no missing values were found, text fields are dimensions, and prices and tax are measures that are averaged.
- **Filters used:** Year of Date, Region, Top 10 and Bottom 10 country filter, a Measure Names filter on the price table, and click-to-filter actions.
- **Calculated fields:** Subsidy Score, Avg Fuel Price and Diesel Volatility.
- **Dashboard design:** nine visualisations.
- **Story design:** [Add the number of visualisations in your Tableau Story].

[Insert the performance test screenshots]

7. Results

7.1 Output Screenshots

[Insert screenshots of the nine visualisations, the full dashboard and the Tableau story]

Key findings from the data

- Countries with Low subsidies average about $3.09 per litre across the three fuels, against about $0.13 at the Very High subsidy level.
- Low-subsidy countries tax fuel at about 37.5% on average, while Medium, High and Very High countries are all near 15%.
- Rwanda, New Zealand, Germany, Turkey and Denmark have the highest average tax rates, at about 39%.
- High-income countries tax fuel at about 34.7%, compared with about 22.9% for Low-income and 21.1% for Middle-income countries.
- High-income countries pay about $3.55 per litre for petrol, against about $1.39 in Low-income countries.
- Average diesel price rose from about $1.39 per litre in 2020 to about $2.47 in 2026.
- Hong Kong ($5.10), Ireland ($4.55) and Norway ($4.43) are the costliest diesel markets; Venezuela, Iran and Libya are the cheapest.
- The Middle East shows the highest diesel volatility and Oceania the lowest. This measure reflects price differences between countries within a region as well as changes over time.

8. Advantages & Disadvantages

Advantages

- One dashboard answers the questions of three different stakeholders.
- Interactive filters and click-to-filter make exploration easy without technical skills.
- The forecast and volatility views support procurement planning.
- The project uses a single CSV, so it is simple to update and share.

Disadvantages

- The data is static, so the dashboard does not update automatically.
- Subsidy level is a category (Low to Very High), not an amount, which limits precise analysis.

- Tax percentage varies widely from week to week, so averages are used to keep charts readable.
- The volatility heat map mixes differences between countries with changes over time.
- Patterns shown are associations and do not prove cause and effect.

9. Conclusion

The dashboard shows that fuel prices, taxes and subsidies differ strongly by region and income level. Low-subsidy and high-income countries have the highest prices, and tax is highest in Europe and Oceania. Diesel prices have risen since 2020, and risk differs by region. Each stakeholder can find a direct answer in their own section of the dashboard, which makes the data easier to use for policy, procurement and research decisions.

10. Future Scope

- Connect a live or scheduled data source so the dashboard refreshes automatically.
- Add more indicators such as GDP, emissions and vehicle counts to explain price and tax differences.
- Add predictive models and alerts for price spikes.
- Move the data to a cloud database and host on a private server with role-based access.
- Improve the mobile layout of the dashboard.